

Rooting the Social Graph with Personal Network Applications

The Social Network Health Project



Rich Bodo

DWeb Camp Berlin 2026 · Hackers Lab

socialnetwork.health · talk ~20 min, then we work

1 · Me

- Sysadmin, developer, manager — '80s through 2010s
- Early decentralized research with **Mozilla** and the **World Food Programme**
- Always an entrepreneur — no big exits — pro-bono help for a half-dozen startups and impact orgs a month
- Pandemic years: learned from a U of R trainer → recognized **social network health** as my highest-impact research area
- DWeb 2024: presented learnings — asked if software could help. *Didn't think so.*

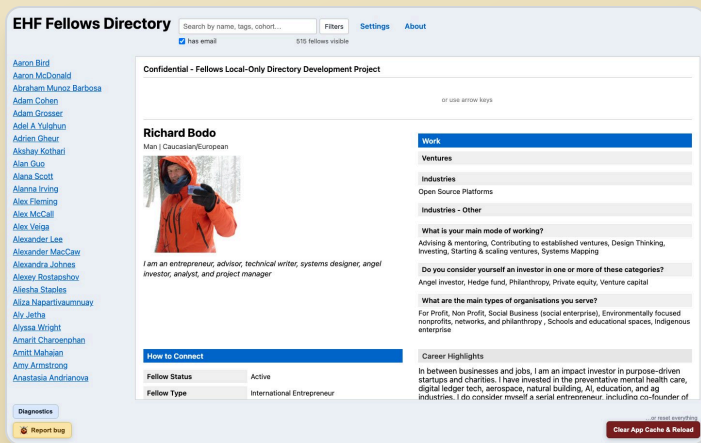


2 · Help-Seeking Impedance

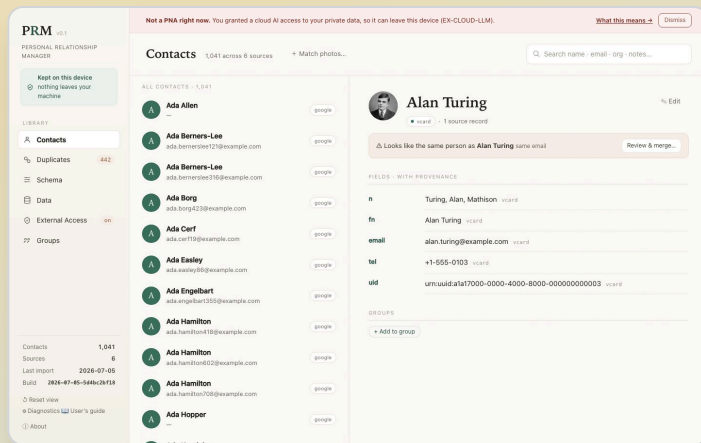
- Changed my tune when **I needed help myself**
- Even a *small* impedance to help-seeking can be too great at times
- Taking the data offline helps — but relationship data is **as sensitive as it gets** for an individual
- I want my communities to seek help **without impedance** — and we need a safe way to do that *outside of SaaS*
- Starting where I am: my egocentric network, building for that group only
- Assumption: people will build for themselves with AI — so I tooled up for that, too

3 · Home-Cooked Meals

- **2025:** built generic local-first software to compete with centralized — *not useful*
- **This year:** home-cooked meals for the egocentric graph — *that works*, and it opens up possibilities



Exit: orgs shut down, or we leave — a snapshot of directory data, out of SaaS.
I'm more connected; so are many others.



Interop: PRM — my tool, also made for one other DWeb camper.

4 · Social Network Health — Project Structure

Data Safety

PNA Toolkit

spec · contracts ·
conformance · reference
designs

Education

SNHDB · Wiki · Videos

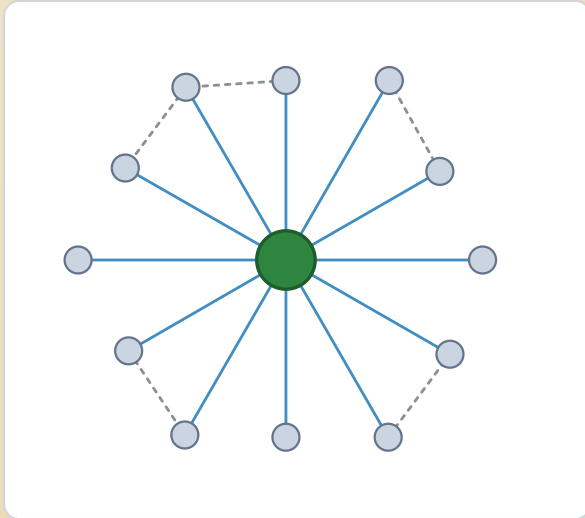
research library, integrated
and updated

Research

Forward-looking protocols

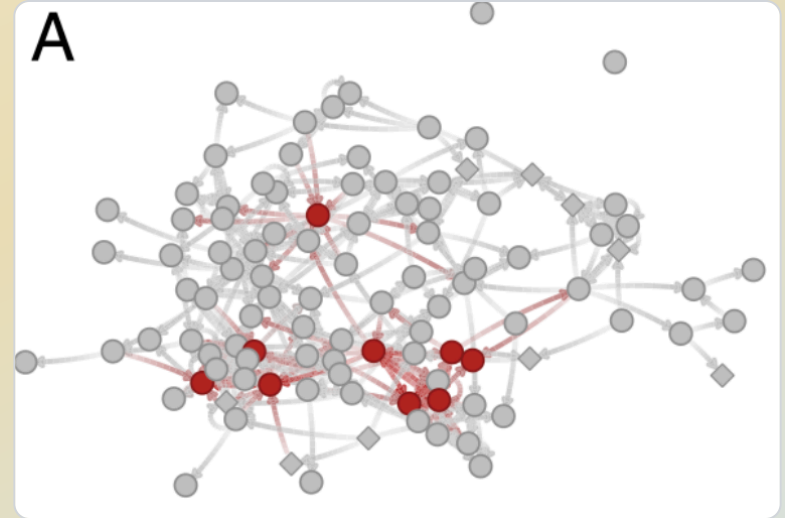
check-in · notification ·
measurement

5 · Definitions



Egocentric graph — you, your direct ties, and what *you know* about who knows whom (dotted)

Ecological validity — findings that hold in real communities, not just in the lab



Community sociograph — the whole community's tie structure, from a real study

6 · Data Safety: The PNA Toolkit — Four Goals, Plainly

1. **Your people stay on your own device** — your contacts and private notes live with you, whatever happens to any online account
2. **You can check that it's telling the truth** — you (or someone helping you) can confirm the app does what it promises
3. **Nothing leaves without you** — nothing goes out unless you choose it, understand the risk, and see exactly what's sent
4. **You won't lose it** — your private notes survive new phones, browser clean-ups, and app updates

Set up Claude Desktop integration

This easy path is for **Chrome and Chrome-derived browsers** (Edge, Brave, Arc). Safari, Firefox, and developers integrating other tools should use the [manual walkthrough](#).

Heads up. You haven't set up a private data folder yet. We recommend *Settings* → *Private data folder* → *Choose folder...* first (takes about ten seconds). Without it, you'll need to redo this setup every time you change a group.

Before you connect a cloud AI — please read

1. **You're leaving the local-only model.** Local AI is hard for most people to run, and Claude Desktop is the option nearly everyone actually wants. But connecting it sends your fellows data — potentially including your private groups and notes — to a SaaS vendor (Anthropic). No one can guarantee what a SaaS vendor will or won't do with data you send it. This breaks the central promise of a personal-network app: that it never talks to a SaaS server. You have to be OK with that to continue.
2. **MCP and LLMs are new, and can misbehave.** MCP integrations are very new. We wrote these extensions to do only benign, read-only things, and the code is auditable — but an LLM driving them can still make mistakes or hit bugs. You accept the risk that something could go wrong with your fellows database. The good news: these extensions only touch two files, and both are recoverable. You can always re-download the shared directory, and you can restore your private data (groups, notes) from a backup or export. So the worst case is recoverable — but this still isn't the spirit of a local-only app, and you have to

Accepting raises the EX-CLOUD-LLM exception and takes this app out of PNA (local-only) mode. [Read the full explanation](#) (opens the explainer and closes this dialog). It is reversible — you can return to PNA mode later.

☐ I understand and accept these risks.

Scroll to the end of the text above to enable the checkbox.

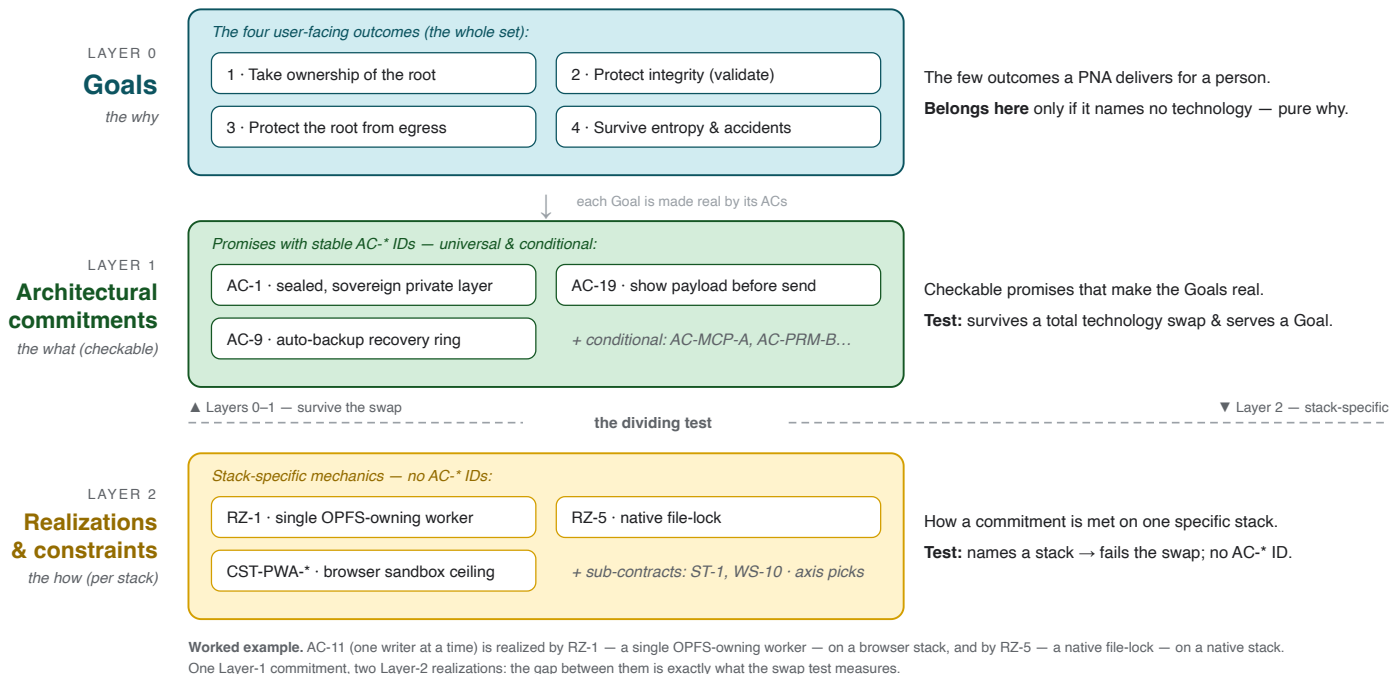
Installing will grant this extension access to everything on your computer.

Goal 3 in the flesh — the consent gate before a cloud AI connects

7 · Data Safety: The Layers

The PNA Spec in three layers

Goals → architectural commitments → realizations · the dividing test: does it survive a total technology swap?



The dividing test: does it survive a total technology swap? (PNA Spec § How the pieces fit together)

8 · Data Safety: Commitments & Reference Designs

Architectural commitments

- **Checkable promises with stable IDs** (AC-1 , AC-19 , ...) — the unit of conformance
- Technology-independent: each survives a total technology swap
- Every typed contract names the AC(s) it realizes; a design attests **AC by AC**

Reference designs

- **fellows_local_db** —  complete, in use · 515 fellows, local-only
- **PRM** —  in progress · personal relationship manager on fixture data
- They prove the constraints are **livable** — made *for* specific people, rapidly

9 · Why Home-Cooked Local-First

- **Safety** — personal data never leaves local user control without full understanding and permission
- **Usability** — scale it down to *one person* and near-perfect feedback and customization become possible.
Feedback drives everything.
- **Simplicity** — a home to protect your data and analyze threats will always be needed; adding complexity there doesn't help. Restrict functionality. The complexity of sync, auth, and sharing is **offloaded** — credible interop is often simpler than that.

10 · Exit or Interop?

- This is a **Facebook contact-data export**
- They give you **names — and nothing else**: no email, no phone, no URL, no id
- Self-evidently not fully-fleshed-out contact data
- Do we need to *sync* with them? Or just **export it and build locally** — take control and flesh it out ourselves

your_friends.json — Facebook "Download Your Information"

```
{  
  "friends": [  
    { "name": "Ada Lovelace", "timestamp": 1576022400 },  
    { "name": "John Doe", "timestamp": 1645392000 },  
    { "name": "Jane Smith", "timestamp": 1678915200 },  
    { "name": "José Ñoño", "timestamp": 1597017600 },  
    { "name": "田中 太郎", "timestamp": 1612137600 },  
    { "name": "Mary O'Brien", "timestamp": 1500000000 }  
  ]  
}
```

That's the whole record. Every field of it.

11 · Methodology

- **Conformance as discipline** — crazy to do in 2025; now it's crazy *not* to. Build context and test constantly.
- **Threat modelling & positioning** — we can afford to refresh these *often* now
- **Encourage user-development** — skills ship with tools; spec versions ship with the toolkit
- **Education is central** — users need to know what they're getting into (AI, OS automation...). We need to know what we're doing: a research library is integrated and updated for devs — *and users will be devs*

Conformance Report — EHF Fellows Local Directory

What this is. A check of whether this repository honestly conforms to the **Personal Network Application (PNA)** spec — a local-first, private-by-default app that mirrors contact data into a user-owned workspace with no remote authority. The spec and this conformance method come from the **Personal Network Toolkit (PNT)**. This report is generated by `scripts/conformance_report.py` (serializing `scripts/conformance_live.py`). It is the **deterministic** layer — it verifies that every `conformance` claim in this app's Security Target (`docs/architecture.ad`) is backed by live, executable evidence. It does not run the spec's LLM/human evaluate flow.

Why it matters. PNA conformance is the spec's proxy for **safe to install** — a local-first app with no remote authority over your data.

Conditionally conformant to the PNA Spec (PNT 0.2)

Every conformance claim this app makes is backed by live, executable evidence, and all deferrals are disciplined. This is the **deterministic** layer of conformance — a full determination also needs the spec's evaluate flow (below).

Conformance is checked in layers, not awarded as a badge:

- **Conformant** — deterministic ✓ · LLM evaluate ✓ · human review ✓ — the goal; not grantable by this report
- **Conditionally conformant** — deterministic evidence verified; evaluate flow pending — **this report**
- **Not conformant** — a `conformance` claim lacks live evidence

→ To complete the determination (● → ●), run the spec's audit flow: **PNT User's Guide** → **Audit a candidate PNA**.

Generated 2026-07-04T11:51:46Z for: 0329fca · Source of truth: `docs/architecture.ad`.

What the IDs mean — **AC** Architectural Commitment (a rule every safe PNA honors) · **CST** Constraint (a platform limit handled honestly, not hidden) · **EX** Exception (a declared departure from PNA rules).

- Conformant rows: 37 of 41
- Deferrals: 0 of 3 max ✓
- Findings: 0 ✓

Deferrals (strict-xfail)

None — zero deferred invariants.

Findings

None. Every `conformance` row cites live, non-deferred evidence; every deferral is anchored and under cap.

Attestation rows

Each row's ID links to its definition in the PNT spec. Status is summarized; each row's realization prose is expandable in the **appendix below**, and the source of live data is `docs/architecture.ad`.

fellows_local_db conformance report — evidence-linked

12 · Methodology Finding: Usability *Is* User Safety

- Users won't use an app they consider unusable — **exceptions to safety are the reality**
- So: develop **countermeasures** and deal with it — this improves the toolkit *and* the spec (the AI exceptions)
- Keep building **with users** — External Access Handling came from exactly this
- And the payoff: users actually *use* the app — **so they can seek help more easily**

13 · The AI Exception: What We Had to Build

"Honesty scales; purity doesn't."

- **Consent gate, fail-closed** — a human action on a surface the app controls; named risks, scroll-to-enable
- **Sealed by default / data-floor** — every field unreadable until *you* mark it shareable; the most sensitive never reach the AI surface
- **Surface minimization** — the cloud-facing surface exposes only the tools the task needs
- **Blast-radius preview** — which fields × how many contacts, *before* you consent
- **Mode banners** — amber "local AI on" (with its honest caveat) · red "**Not a PNA right now**"
- **Per-read approval** — each AI read held for your Approve / Deny, per contact
- **Exceptions ledger** — every grant itemized: which contact, which fields, when approved, when last read — each with its own **Revoke**
- **Return to PNA mode** — one action drops every grant; the next read is withheld
- **Staged writes** — the AI proposes; a human commits. *No promote tool exists on the AI surface*
- **Honest non-claims** — sent data can't be recalled; the consuming model can't be identified. The UI says so.

*"The surest way to fail at protecting data is to ship a tool too pure to use. The fellows users demanded a cloud LLM. ... In return it provides real productivity. **That is the equation.**"*

14 · Methodology Finding: Ongoing Processes

- **Positioning updates found siblings from other eras** — the survey isn't a one-time artifact; refreshing it keeps surfacing kin projects
- **Validating other apps improves the spec** — running the three layers against *Signal* sharpened the commitments

15 · A Longer-Term Plan

Presented here first.

1. **A safe place for human relationship data** — validated tools that bring contact & relationship data local, so we can interact outside SaaS with confidence. Start with the simplest things that work: *home-cooked meals*.
2. **Egocentric analysis** — validated tools that bring *communications* data local, to understand our egocentric networks and act to improve outcomes within them
3. **Sociocentric analysis & interaction** — validated *protocols* that may let remote communities practice what was once only local, practitioner-guided work. Advanced tech enters here — **research teams included before interventions.**

16 · Notification Protocol — Connection Requests with Staged Disclosure

One-button help-seeking — disclosing as little as possible, to as few as possible, at every step. Staged disclosure is the friction reduction.

Actors

- **Requester** — the person seeking a 1:1
- **Responder** — an audience member who answers a ping
- **Audience** — *any addressable set of helpers*: a PRM tag group ↔ a **super contact**
- **Coordinator** — holds staged state, matches while both sides stay blinded
- **Steward** — accountable human: proposes the super contact, operates the coordinator

S0 · Compose (local) — audience · coarse topic · time window. Nothing leaves the device.



S1 · Availability ping — "someone would like a 1:1, window X." No topic, no identity.



S2 · Topic gate — coarse topic → *available responders only*. The wider audience never learns it.



S3 · Match + mutual consent — identities cross only now. Reveal mode is a per-audience policy knob.



S4 · Close-out — warm thanks to the unchosen · no-match → **pre-generated fallbacks, local**.

"The fallback path is entirely local — zero privacy cost at the moment of maximum vulnerability."

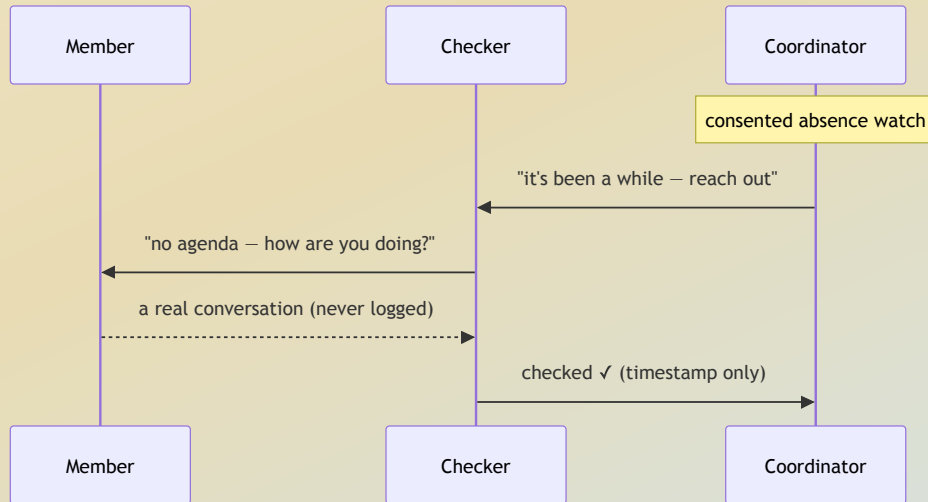
17 · Check-in Protocol — Proactive Peer Check-ins

A help-seeking button cannot reach someone who has stopped reaching.

Actors

- **Member** — anyone in the community
- **Checker** — *named by the member* (the phone tree); never assigned
- **Steward** — accountable human; escalations + crisis resources
- **Coordinator** — optional software; pairings + timestamps, **never content**

Stages — 1 human-human cadence (no software) → 2 triggered (consented absence / event signals) → 3 escalation paths. *Least risky first; each gates the next.*



Stage 2, success case — the nudge carries elapsed time only, no inference; content never touches the system

Not a cron job — every choice is research-consulted: top-down checker assignment overlaps only **13–23%** with network-optimal picks (Pickering, Wyman et al. 2022; Wyman 2010/2019).

18 · Community Network Health from Encrypted Egocentric Data

The theory already exists. Smith 2012, "*Macrostructure from Microstructure*": whole-network structure can be **estimated from egocentric samples alone** — the fragments each of us can safely hold.

- This one is a **thesis, not yet a protocol**: *these calculations can be made without anyone surrendering their graph*
- The engine is an **ERGM** — at heart, **logistic regression**: given local tendencies (clustering, homophily, sparsity), how likely is *this tie* vs. not?
- The payoff, implemented on privacy-preserving substrates: **community-level aggregate health metrics** — cohesion · reach · isolation — *with honest uncertainty ranges, never individual scores*

Sample — consented egocentric fragments: who-knows-whom, nothing more



Fit — an ERGM via logistic regression: which tie patterns are likelier in *this* community



Simulate — generate many whole networks consistent with the fitted model, scaled to the full population



Quantify — the spread across those simulations → a *range*, not a false point estimate



Derive — aggregate community-health metrics + uncertainty

"Gate more, model more" — privacy gating and generative reconstruction are the same coin.

19 · Demos

- **Validation** — the conformance suite running against a real app
- **SNHDB research queries** — asking the research corpus a question, with citations
- **Paper resolver** — DOI/topic → metadata + legal open-access full text

Pre-recorded video — narrated live.

20 · Workshop Options

Break out. We can workshop:

- **A.** Critiques of the plan, the PNA Toolkit and spec — areas to improve
- **B.** SNHDB research queries
- **C.** Validation of *your* favorite app
- **D.** One-shot a SaaS backup app

...but we meet back and synthesize for the last 15 minutes.

socialnetwork.health · discuss@socialnetwork.health

21 · Do It Yourself

Validate an app

```
github.com/social-network-health/  
personal_network_toolkit
```

1. Clone — the **pna-toolkit skill** ships with it
2. In Claude Code: "*evaluate <app> against the PNA spec*" → the skill runs the **evaluate flow** → a typed `evaluate-report.json`
3. `just view-reports` — every validation, rendered; the casebook lives in `evaluations/` (e.g. Signal Desktop)

Ask the research

```
github.com/social-network-health/  
snhdb
```

1. Clone (~1 GB, all local) — install the **/snhdb skill** (one symlink)
2. In any Claude Code session:

/snhdb What does the research say about loneliness interventions for remote workers? Cite papers.

The cards are the index; answers cite the papers.

Critique the spec & plan

The PNA Spec — goals, commitments, exceptions:

```
personal_network_toolkit/  
spec/PNA_Spec.md
```

The plan — the three steps, and the research behind each:

```
social-network-health/  
research/plan.md
```

Issues and PRs welcome — it's all *checked, not awarded*.